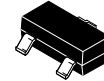


Digital FET, P-Channel

FDV304P, FDV304P-F169


SOT-23-3
CASE 318-08

General Description

This P-Channel enhancement mode field effect transistors is produced using onsemi's proprietary, high cell density, DMOS technology. This very high density process is tailored to minimize on-state resistance at low gate drive conditions. This device is designed especially for application in battery power applications such as notebook computers and cellular phones. This device has excellent on-state resistance even at gate drive voltages as low as 2.5 V.

Features

- -25 V, -0.46 A Continuous, -1.5 A Peak
 - ♦ $R_{DS(on)} = 1.1 \Omega @ V_{GS} = -4.5 V$
 - ♦ $R_{DS(on)} = 1.5 \Omega @ V_{GS} = -2.7 V$
- Very Low Level Gate Drive Requirements Allowing Direct Operation in 3 V Circuits. $V_{GS(th)} < 1.5 V$
- Gate-Source Zener for ESD Ruggedness. > 6 kV Human Body Model
- Compact Industry Standard SOT-23 Surface Mount Package
- This Device is Pb-Free and Halide Free

ABSOLUTE MAXIMUM RATINGS ($T_A = 25^\circ C$ unless otherwise noted.)

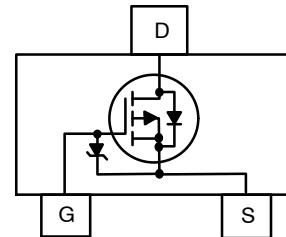
Symbol	Parameter	Value	Unit
V_{DSS}	Drain-Source Voltage	-25	V
V_{GSS}	Gate-Source Voltage	-8	V
I_D	Drain Current - Continuous	-0.46	A
	Drain Current - Pulsed	-1.5	
P_D	Maximum Power Dissipation	0.35	W
T_J, T_{STG}	Operating and Storage Temperature Range	-55 to 150	$^\circ C$
ESD	Electrostatic Discharge Rating MIL-STD-883D Human Body Model (100 pF/1500 Ω)	6.0	kV

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

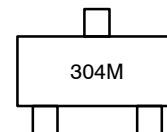
THERMAL CHARACTERISTICS ($T_A = 25^\circ C$ unless otherwise noted.)

Symbol	Parameter	Value	Unit
$R_{\theta JA}$	Thermal Resistance, Junction-to-Ambient	357	$^\circ C/W$

ELECTRICAL CONNECTION



MARKING DIAGRAM



304 = Specific Device Code
M = Date Code

ORDERING INFORMATION

Device	Package	Shipping [†]
FDV304P	SOT-23-3 (Pb-Free, Halide-Free)	3000 / Tape & Reel
FDV304P-F169	SOT-23-3 (Pb-Free, Halide-Free)	3000 / Tape & Reel

[†]For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specification Brochure, [BRD8011/D](#).

FDV304P, FDV304P-F169

ELECTRICAL CHARACTERISTICS (T_A = 25°C unless otherwise noted.)

Symbol	Parameter	Test Conditions	Min	Typ	Max	Unit
--------	-----------	-----------------	-----	-----	-----	------

OFF CHARACTERISTICS

BV _{DSS}	Drain-Source Breakdown Voltage	V _{GS} = 0 V, I _D = -250 μA	-25	-	-	V
ΔBV _{DSS} /ΔT _J	Breakdown Voltage Temp. Coefficient	I _D = -250 μA, Referenced to 25°C	-	-22	-	mV/°C
I _{DSS}	Zero Gate Voltage Drain Current	V _{DS} = -20 V, V _{GS} = 0 V	-	-	-1	μA
		V _{DS} = -20 V, V _{GS} = 0 V, T _J = 55°C	-	-	-10	
I _{GSS}	Gate-Body Leakage Current	V _{GS} = -8 V, V _{DS} = 0 V	-	-	-100	nA

ON CHARACTERISTICS (Note 1)

ΔV _{GS(th)} /ΔT _J	Gate Threshold Voltage Temp. Coefficient	I _D = -250 μA, Referenced to 25°C	-	2.1	-	mV/°C
V _{GS(th)}	Gate Threshold Voltage	V _{DS} = V _{GS} , I _D = -250 μA	-0.65	-0.86	-1.5	V
R _{DS(on)}	Static Drain-Source On-Resistance	V _{GS} = -2.7 V, I _D = -0.25 A	-	1.22	1.5	Ω
		V _{GS} = -4.5 V, I _D = -0.5 A	-	0.87	1.1	
		V _{GS} = -4.5 V, I _D = -0.5 A, T _J = 125°C	-	1.21	2	
I _{D(on)}	On-State Drain Current	V _{GS} = -2.7 V, V _{DS} = -5 V	-0.5	-	-	A
		V _{GS} = -4.5 V, V _{DS} = -5 V	-1	-	-	
g _{FS}	Forward Transconductance	V _{DS} = -5 V, I _D = -0.5 A	-	0.8	-	S

DYNAMIC CHARACTERISTICS

C _{iss}	Input Capacitance	V _{DS} = -10 V, V _{GS} = 0 V, f = 1.0 MHz	-	63	-	pF
C _{oss}	Output Capacitance		-	34	-	
C _{rss}	Reverse Transfer Capacitance		-	10	-	

SWITCHING CHARACTERISTICS (Note 1)

t _{D(on)}	Turn-On Delay Time	V _{DD} = -6 V, I _D = -0.5 A, V _{GS} = -4.5 V, R _{GEN} = 50 Ω	-	7	20	ns
t _r	Turn-On Rise Time		-	8	20	
t _{D(off)}	Turn-Off Delay Time		-	55	110	
t _f	Turn-Off Fall Time		-	35	70	
Q _g	Total Gate Charge	V _{DS} = -5 V, I _D = -0.25 A, V _{GS} = -4.5 V	-	1.1	1.5	nC
Q _{gs}	Gate-Source Charge		-	0.32	-	
Q _{gd}	Gate-Drain Charge		-	0.25	-	

DRAIN-SOURCE DIODE CHARACTERISTICS AND MAXIMUM RATINGS

I _S	Maximum Continuous Drain–Source Diode Forward Current		–	–	–0.5	A
V _{SD}	Drain–Source Diode Forward Voltage	V _{GS} = 0 V, I _S = –0.5 A (Note 1)	–	–0.89	–1.2	V

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

1. Pulse Test: Pulse Width ≤ 300 μs, Duty Cycle ≤ 2.0%.

TYPICAL CHARACTERISTICS

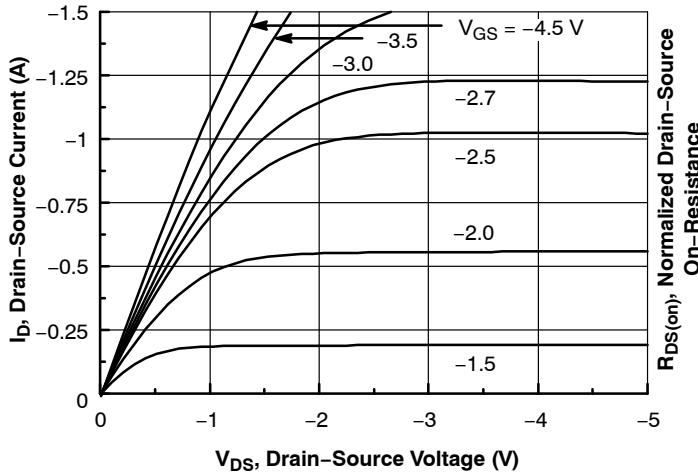


Figure 1. On-Region Characteristics

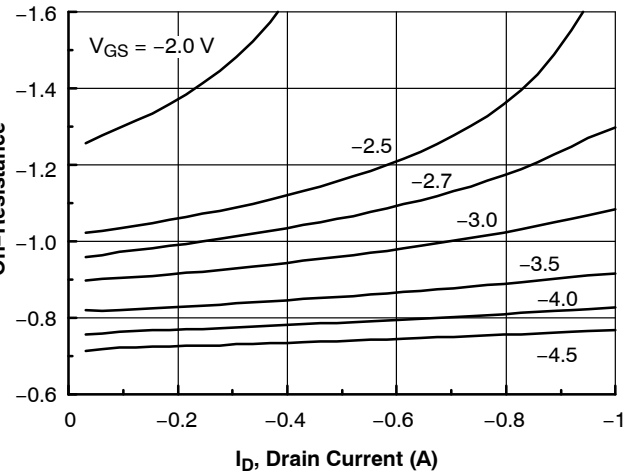


Figure 2. On-Resistance Variation with Drain Current and Gate Voltage

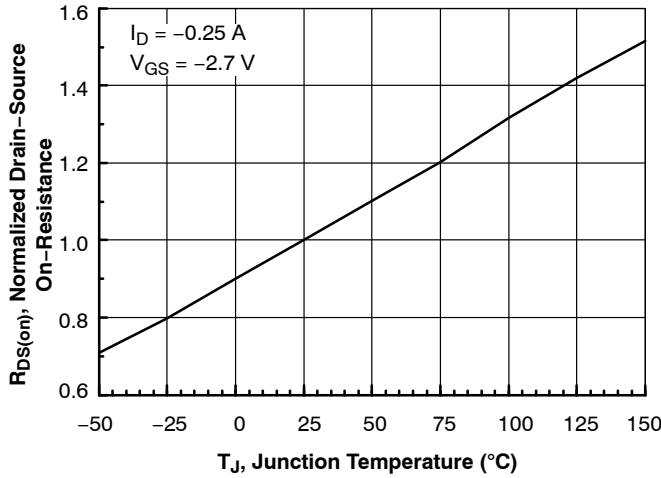


Figure 3. On-Resistance Variation with Temperature

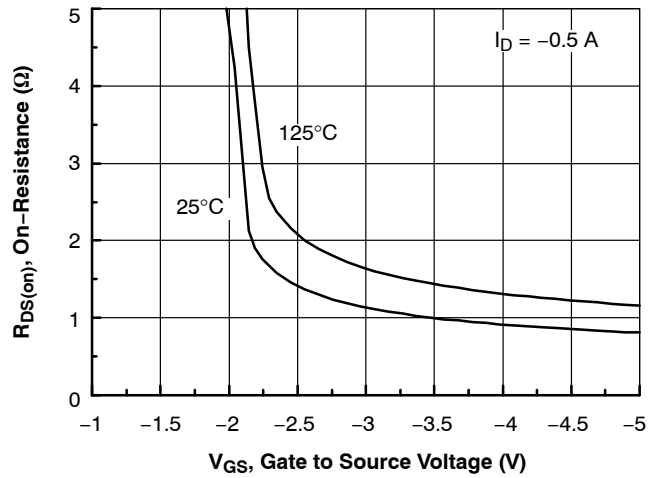


Figure 4. On Resistance Variation with Gate-To-Source Voltage

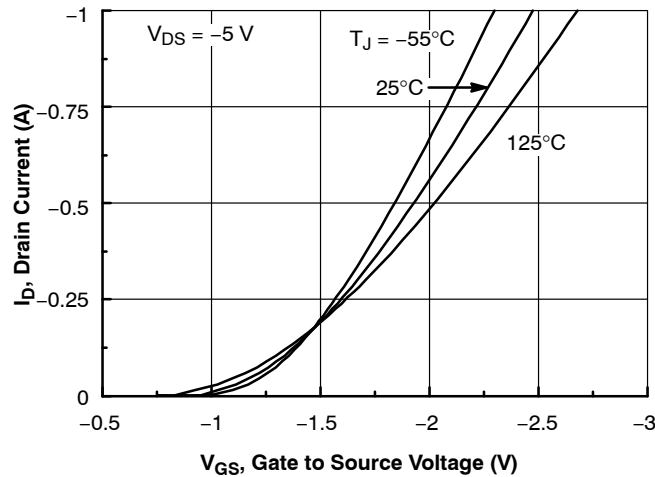


Figure 5. Transfer Characteristics

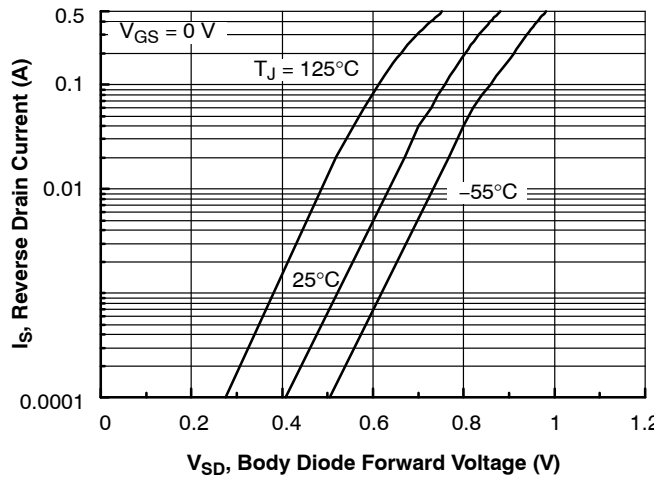


Figure 6. Body Diode Forward Voltage Variation with Source Current and Temperature

TYPICAL CHARACTERISTICS (continued)

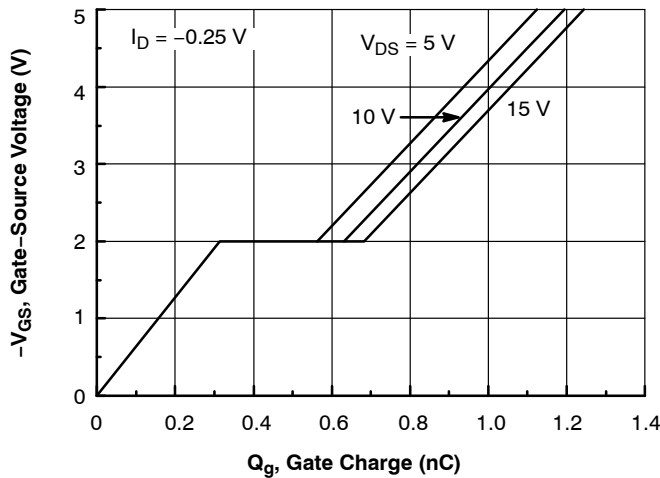


Figure 7. Gate Charge Characteristics

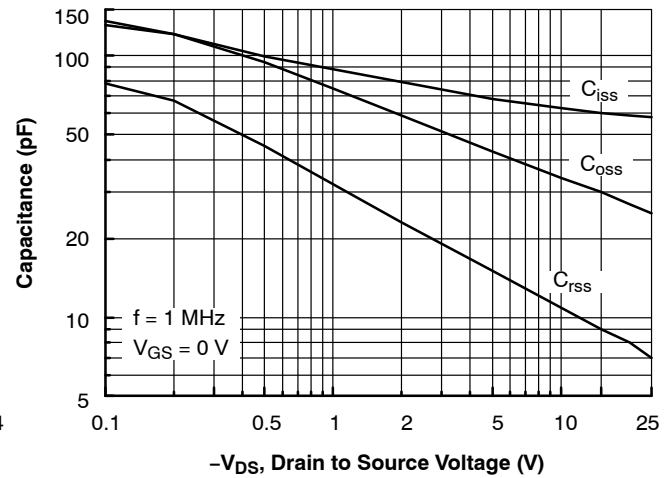


Figure 8. Capacitance Characteristics

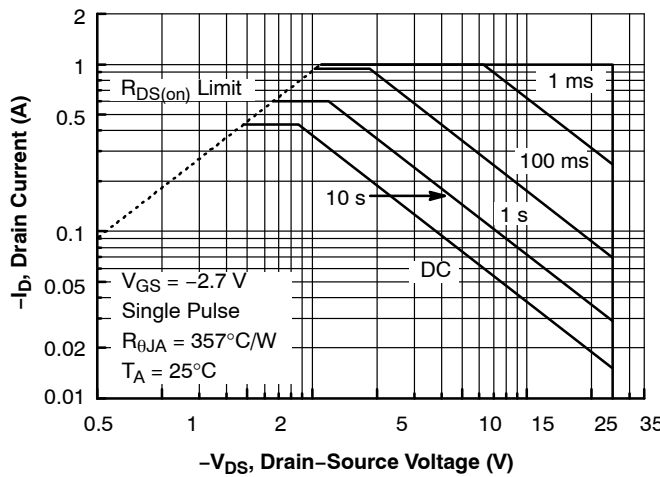


Figure 9. Maximum Safe Operating Area

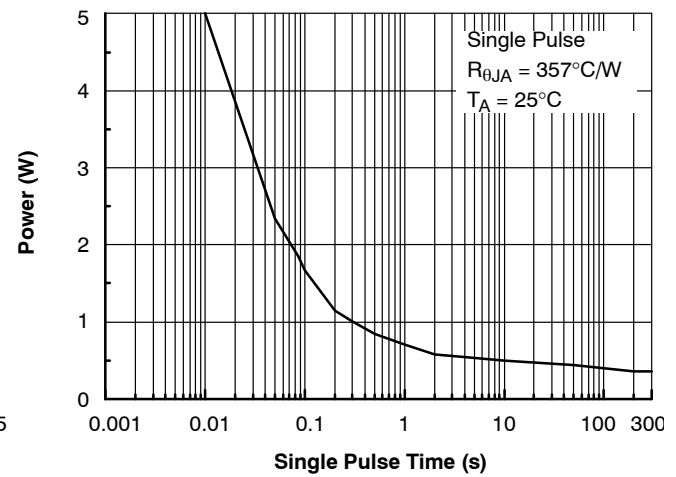


Figure 10. Maximum Pulse Maximum Power Dissipation

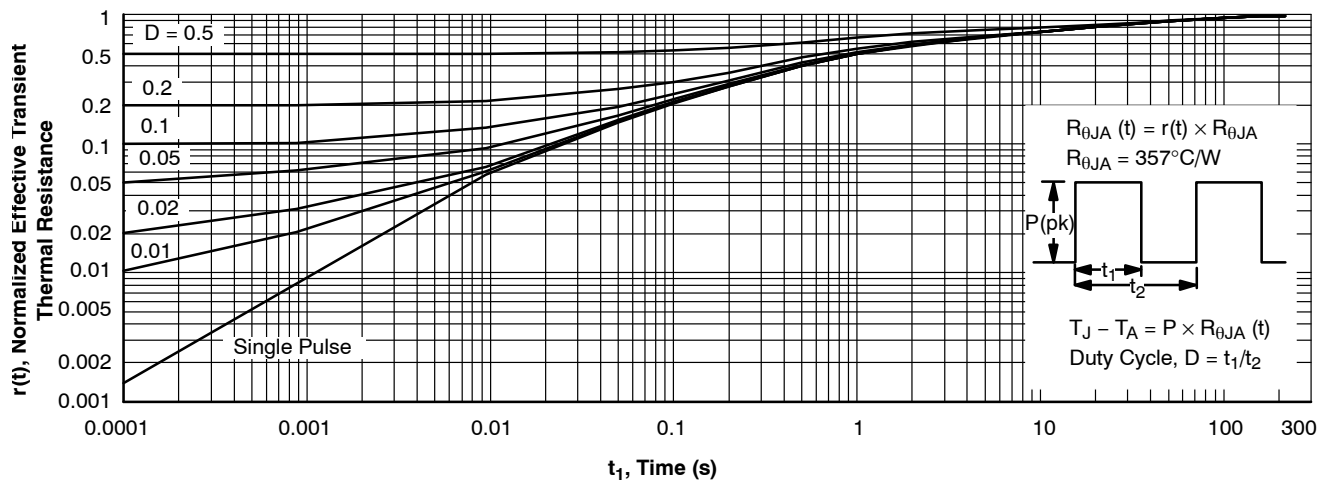
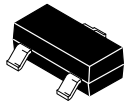


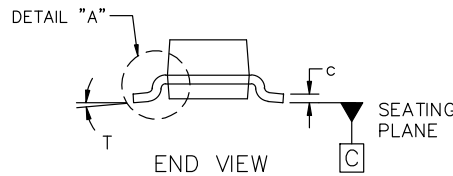
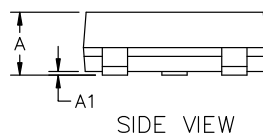
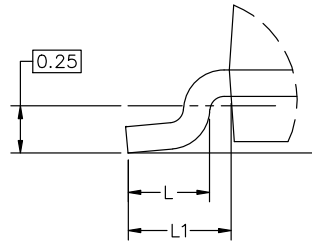
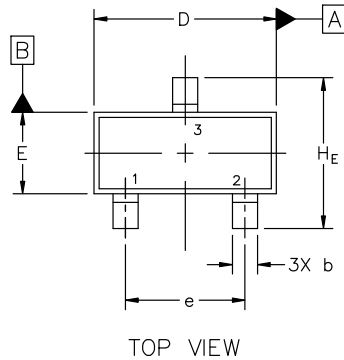
Figure 11. Transient Thermal Response Curve



SCALE 4:1

SOT-23 (TO-236) 2.90x1.30x1.00 1.90P
CASE 318
ISSUE AU

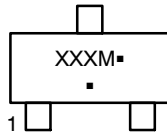
DATE 14 AUG 2024



MILLIMETERS			
DIM	MIN	NOM	MAX
A	0.89	1.00	1.11
A1	0.01	0.06	0.10
b	0.37	0.44	0.50
c	0.08	0.14	0.20
D	2.80	2.90	3.04
E	1.20	1.30	1.40
e	1.78	1.90	2.04
L	0.30	0.43	0.55
L1	0.35	0.54	0.69
HE	2.10	2.40	2.64
T	0°	---	10°

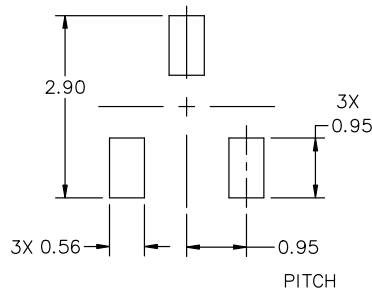
NOTES:

1. DIMENSIONING AND TOLERANCING PER ASME Y14.5M, 2018.
2. CONTROLLING DIMENSIONS: MILLIMETERS.
3. MAXIMUM LEAD THICKNESS INCLUDES LEAD FINISH. MINIMUM LEAD THICKNESS IS THE MINIMUM THICKNESS OF THE BASE MATERIAL.
4. DIMENSIONS D AND E DO NOT INCLUDE MOLD FLASH, PROTRUSIONS, OR GATE BURRS.

GENERIC MARKING DIAGRAM*


XXX = Specific Device Code
M = Date Code
▪ = Pb-Free Package

*This information is generic. Please refer to device data sheet for actual part marking. Pb-Free indicator, "G" or microdot "▪", may or may not be present. Some products may not follow the Generic Marking.


RECOMMENDED MOUNTING FOOTPRINT

* For additional information on our Pb-Free strategy and soldering details, please download the onsemi Soldering and Mounting Techniques Reference Manual, SOLDERM/D.

STYLES ON PAGE 2

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DATE 14 AUG 2024

STYLE 1 THRU 5: CANCELLED	STYLE 6: PIN 1. BASE 2. EMITTER 3. COLLECTOR	STYLE 7: PIN 1. EMITTER 2. BASE 3. COLLECTOR	STYLE 8: PIN 1. ANODE 2. NO CONNECTION 3. CATHODE		
STYLE 9: PIN 1. ANODE 2. ANODE 3. CATHODE	STYLE 10: PIN 1. DRAIN 2. SOURCE 3. GATE	STYLE 11: PIN 1. ANODE 2. CATHODE 3. CATHODE-ANODE	STYLE 12: PIN 1. CATHODE 2. CATHODE 3. ANODE	STYLE 13: PIN 1. SOURCE 2. DRAIN 3. GATE	STYLE 14: PIN 1. CATHODE 2. GATE 3. ANODE
STYLE 15: PIN 1. GATE 2. CATHODE 3. ANODE	STYLE 16: PIN 1. ANODE 2. CATHODE 3. CATHODE	STYLE 17: PIN 1. NO CONNECTION 2. ANODE 3. CATHODE	STYLE 18: PIN 1. NO CONNECTION 2. CATHODE 3. ANODE	STYLE 19: PIN 1. CATHODE 2. ANODE 3. CATHODE-ANODE	STYLE 20: PIN 1. CATHODE 2. ANODE 3. GATE
STYLE 21: PIN 1. GATE 2. SOURCE 3. DRAIN	STYLE 22: PIN 1. RETURN 2. OUTPUT 3. INPUT	STYLE 23: PIN 1. ANODE 2. ANODE 3. CATHODE	STYLE 24: PIN 1. GATE 2. DRAIN 3. SOURCE	STYLE 25: PIN 1. ANODE 2. CATHODE 3. GATE	STYLE 26: PIN 1. CATHODE 2. ANODE 3. NO CONNECTION
STYLE 27: PIN 1. CATHODE 2. CATHODE 3. CATHODE	STYLE 28: PIN 1. ANODE 2. ANODE 3. ANODE				

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